The Box Principle — problem 4.73

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The Sikinian road system is such that three roads meet at each intersection. Prove the following property of the Sikinian road system : Start at any intersection A_1, and drive along any of the three roads to the next intersection A_2. At A_2 turn right and go to the next intersection A_3. At A_3 turn left, and so on, turning left and right alternately. Then you will eventually return to your starting point A_1.